# FAANG Computer Networks Interview Master Sheet

## How to Use

- Prepare each question in 3 levels: 30 sec answer, 2 min detailed answer, deep dive with examples.
- For every concept know: definition, why needed, working, real-world example, advantages, limitations.
- Focus on OSI/TCP-IP, routing, protocols, troubleshooting, security, and scenario-based design.

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# 1. Basics of Networking (Easy)

1. What is a computer network?
2. Why do we need networking?
3. Types of networks: LAN, MAN, WAN, PAN.
4. What is a node, link, bandwidth, latency, throughput?
5. What is network topology?
6. Types of topology: Bus, Star, Ring, Mesh, Tree.
7. Advantages and disadvantages of Star topology.
8. What is simplex, half duplex, full duplex?
9. Guided vs unguided media.
10. Copper vs Fiber optic cable.
11. What is NIC?
12. What is MAC address?
13. What is IP address?
14. Public IP vs Private IP.
15. Static IP vs Dynamic IP.
16. What is APIPA?
17. What is packet, frame, segment, datagram?

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# 2. OSI Model & TCP/IP Model (Easy to Medium)

1. Explain OSI model in detail.
2. Functions of each OSI layer.
3. Mnemonic to remember OSI layers.
4. Explain TCP/IP model.
5. OSI vs TCP/IP model.
6. Which protocols work at each layer?
7. Which devices work at each layer?
8. Encapsulation and decapsulation.

9. Why layering is needed in networking?
10. Which layer uses ports?
11. Which layer uses IP?
12. Which layer uses MAC?

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# 3. IP Addressing & Subnetting (Medium)

1. What is IPv4?
2. What is IPv6?
3. IPv4 vs IPv6.
4. What is subnet mask?
5. What is subnetting?
6. Why subnetting is needed?
7. CIDR notation.
8. What is default gateway?
9. Network ID vs Host ID.
10. Broadcast address.
11. Private IP ranges.
12. Loopback address.
13. What is NAT?
14. PAT vs NAT.
15. Solve subnetting numericals.
16. How many hosts in /24, /16, /30 etc?

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# 4. Transport Layer (Medium)

1. What is Transport layer?
2. TCP in detail.
3. UDP in detail.
4. TCP vs UDP.
5. Connection-oriented vs Connectionless.
6. Reliable vs Best-effort delivery.
7. What is port number?
8. Well-known ports.
9. What is socket?
10. Flow control.
11. Congestion control.
12. Sliding window protocol.
13. Stop and Wait protocol.
14. Selective Repeat.
15. Go-Back-N ARQ.

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# 5. TCP Deep Dive (Medium to Hard)

1. Explain 3-way handshake.
2. Why 3-way handshake and not 2-way?
3. Explain 4-way termination.
4. TCP flags: SYN, ACK, FIN, RST, PSH, URG.
5. Sequence number and acknowledgment number.
6. What happens if ACK is lost?
7. What happens if SYN is lost?
8. Why TIME_WAIT state exists?
9. What is retransmission timeout?
10. What is MSS?
11. What is window size?
12. What is slow start?
13. What is congestion avoidance?
14. What is fast retransmit?
15. What is fast recovery?
16. Why TCP is slower than UDP?
17. Head-of-line blocking.

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# 6. Routing & Network Layer (Medium to Hard)

1. What is routing?
2. Static routing vs Dynamic routing.
3. What is router?
4. Router vs Switch.
5. Hub vs Switch.
6. Bridge vs Switch.
7. Gateway vs Router.
8. What is ARP?
9. What is ICMP?
10. What does ping command do?
11. What does traceroute do?
12. TTL in IP packet.
13. What is fragmentation?
14. Why fragmentation happens?
15. Distance Vector routing.
16. Link State routing.
17. RIP protocol.
18. OSPF protocol.
19. BGP protocol.
20. Default route.

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## 7. Application Layer Protocols (Easy to Medium)

1. What is HTTP?
2. What is HTTPS?
3. HTTP vs HTTPS.
4. What is DNS?
5. How DNS works?
6. DNS forwarder.
7. What is DHCP?
8. What is SMTP?
9. POP3 vs IMAP.
10. What is FTP?
11. Active vs Passive FTP.
12. What is SSH?
13. What is Telnet?
14. What is REST API?
15. What are cookies and sessions?
16. What is CDN?
17. What is WebSocket?

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## 8. Famous Real-World Questions (Must Prepare)

1. What happens when you type google.com in browser?
2. How does DNS resolution happen?
3. How does browser load a webpage?
4. How does email reach receiver?
5. How does video streaming work?
6. Why buffering happens in videos?
7. How does WhatsApp call work?
8. Why gaming prefers UDP?
9. Why HTTPS needs certificates?
10. Why websites use load balancers?

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## 9. Network Security (Medium to Hard)

1. What is firewall?
2. Types of firewalls.
3. What is VPN?
4. Advantages and disadvantages of VPN.
5. What is SSL/TLS?
6. How HTTPS is secure?
7. What is RSA algorithm?

8. Symmetric vs Asymmetric encryption.
9. What is hashing?
10. What is man-in-the-middle attack?
11. What is DDoS attack?
12. What is phishing?
13. What is IDS vs IPS?
14. What is Zero Trust networking?

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## 10. Performance & Delays (Medium)

1. Types of delays in network.
2. Transmission delay.
3. Propagation delay.
4. Processing delay.
5. Queuing delay.
6. Latency vs Throughput.
7. Jitter.
8. Why packets drop?
9. How congestion occurs?
10. How CDN reduces latency?

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## 11. Wireless & Modern Networking (Medium)

1. How Wi-Fi works?
2. What is SSID?
3. What is WPA/WPA2/WPA3?
4. What is hotspot?
5. What is Bluetooth vs Wi-Fi?
6. What is 4G vs 5G?
7. What is IoT networking?
8. What is SDN?
9. What is Cloud networking?
10. What is VLAN?

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## 12. Troubleshooting & Scenario Questions (Hard)

1. Internet connected but website not opening. Diagnose.
2. Ping works but browser doesn't. Why?
3. Slow network in office. How will you debug?
4. Users can access local server but not internet.
5. DNS server down impact?

6. Why packet loss happens on video call?
7. How would you design scalable chat server?
8. How would you handle millions of requests/sec?
9. Why microservices need service discovery?
10. How does load balancer improve availability?
11. Why TCP connections exhaust server resources?
12. How to secure public APIs?

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# 13. FAANG Hard / Deep Questions

1. Explain congestion control mathematically.
2. How QUIC improves over TCP?
3. HTTP/1.1 vs HTTP/2 vs HTTP/3.
4. Why HTTP/3 uses UDP?
5. CAP theorem relevance in distributed systems.
6. Consistent hashing in load balancing.
7. Anycast vs Unicast vs Multicast.
8. How BGP failure can affect internet globally?
9. How CDN edge caching works internally.
10. Explain reverse proxy vs forward proxy.
11. Kernel bypass networking (DPDK concept).
12. Ephemeral ports and exhaustion.
13. SYN flood attack and mitigation.
14. Why Nagle's algorithm exists?
15. How TCP keepalive works?

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# 14. Top 30 Quick Revision Questions

1. OSI layers.
2. TCP/IP layers.
3. TCP vs UDP.
4. Hub vs Switch.
5. Router vs Gateway.
6. MAC vs IP.
7. IPv4 vs IPv6.
8. DNS.
9. DHCP.
10. NAT.
11. ARP.
12. ICMP.
13. Ping.
14. HTTP vs HTTPS.
15. SMTP.

16. FTP.
17. SSH.
18. 3-way handshake.
19. 4-way close.
20. Ports.
21. Firewall.
22. VPN.
23. Load balancer.
24. CDN.
25. Subnetting.
26. Latency.
27. Throughput.
28. Packet loss.
29. Congestion.
30. What happens when you type google.com.

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# Interview Strategy

- Start with simple definition.
- Explain where used.
- Give packet flow example.
- Mention pros/cons.
- Compare with alternatives.
- Use real systems examples like browser, Netflix, WhatsApp, Google, cloud infra.